

SSC CGL TIER-1 SPEED & ACCURACY BIBLE

THE 10-SECOND SHORTCUTS & ELIMINATION PLAYBOOK

Complete Blueprint for Quantitative Aptitude, Reasoning, English & GA
Reverse-Engineered from Recent TCS Question Papers & Examiner Traps

1. THE EXAMINER'S PSYCHOLOGY & TRAP ARCHITECTURE

SSC CGL questions are deliberately created with predefined structural patterns. Understanding how options are generated allows you to bypass 70% of calculations:

Trap Mechanism	Examiner's Thinking	The 10-Second Counter-Attack
The Camouflage Twins	Creates the correct answer, then adds an inverse or opposite sign to catch calculation errors.	If two options are $+144$ & -144 , or $5/12$ & $12/5$, eliminate the other two options immediately.
The Pi (π) Multiplier	Mensuration formulas require π ($22/7 = 2 \times 11$). Correct answers are multiples of 11.	Test divisibility by 11 (Odd Sum - Even Sum = 0 or 11) and 7 on options directly.
The Extreme Outlier Trap	Adds one very high or low number to make guessing feel random (e.g. 24, 26, 28, 142).	Outliers are almost never correct. Eliminate extremes and choose between the clustered median options.
Symmetric Cancellation	Cyclic algebraic expressions with a, b, c are constructed to balance out completely.	The answer to symmetric polynomial fractions is almost universally 0, 1, or 3 .

2. QUANTITATIVE APTITUDE: 10-SECOND SPEED ARSENAL

■ Divisibility Rules for 72 & 88 (Tail-Head Split)

Examiner Trap / Setup: Question gives a 9-digit number $789x5432y$ divisible by 72, asks for $(2x + 3y)$.

10-Second Method: Split 72 into 8×9 . (1) Check last 3 digits (32y) for div by 8 $\rightarrow$ gives 'y' instantly. (2) Sum all digits for div by 9 $\rightarrow$ gives 'x'.

Live Example: $32y \div 8 = 328 \rightarrow y=8$. Sum = $7+8+9+x+5+4+3+2+8 = 46+x$. Next multiple of 9 is 54 $\rightarrow x=8$. Value = $2(8)+3(8)=40$.

■ Percentage: The AB Product / Fraction Ladder

Examiner Trap / Setup: Price of sugar increases by 25%. By how much must consumption decrease to keep expense constant?

10-Second Method: If increase is $+a/b$, required decrease is ALWAYS $-a/(a+b)$. No equations needed.

Live Example: $+25\% = +1/4$. Required decrease = $1/(4+1) = 1/5 = 20\%$. Solved in 3 seconds.

■ Profit, Loss & MP Formula

Examiner Trap / Setup: A dealer gives 10% discount and still makes 20% profit. If CP = 300, find MP.

10-Second Method: Use ratio $MP/CP = (100 + P\%) / (100 - D\%)$. Direct proportionality.

Live Example: $MP/CP = (100+20)/(100-10) = 120/90 = 4/3$. Since CP is 300 (3×100), $MP = 4 \times 100 = 400$.

■ Simple & Compound Interest: 2-Year & 3-Year Diff

Examiner Trap / Setup: Difference between CI and SI for 2 years at R% on P is given.

10-Second Method: 2 Years: $\text{Diff} = P(R/100)^2$. For 3 Years: $\text{Diff} = P(R/100)^2 \times [(300+R)/100]$.

Live Example: If $P=10,000$, $R=10\%$, 2-yr Diff = $10000 \times (10/100)^2 = \text{Rs } 100$.

■ Time, Speed & Distance: Early vs Late Shortcut

Examiner Trap / Setup: Going at 4 km/h reaches 10 min late; at 5 km/h reaches 5 min early. Find distance.

10-Second Method: Distance = $[S1 \times S2 / |S1 - S2|] \times [\text{Total Time Diff in Hours}]$.

Live Example: $D = [4 \times 5 / (5-4)] \times [(10+5)/60 \text{ hr}] = 20 \times (15/60) = 5 \text{ km}$.

■ Time & Work: 'Leaving Before Completion' Method

Examiner Trap / Setup: A takes 10 days, B takes 15 days. B leaves 2 days BEFORE completion.

10-Second Method: Do NOT let B leave! Add B's 2 days of work to total work, then divide by ($\text{Eff}_A + \text{Eff}_B$).

Live Example: $\text{LCM}(10,15)=30$. $\text{Eff}(A)=3$, $\text{Eff}(B)=2$. B's 2 days = 4 units. Total = $30+4=34$. Days = $34/(3+2) = 6.8$ days.

■ Algebra: Value Putting & Degrees of Freedom

Examiner Trap / Setup: If $a + b + c = 0$, find the value of $(a^2/bc) + (b^2/ca) + (c^2/ab)$.

10-Second Method: More variables than equations? Put $a=1$, $b=1$, $c=-2$ (Sum=0). Calculate directly.

Live Example: Expression = $(1/-2) + (1/-2) + (4/1) = -0.5 - 0.5 + 4 = 3$. Instant solution.

■ Trigonometry: The 0° / 45° / 90° Rule

Examiner Trap / Setup: Simplify $(\sin \theta / 1 + \cos \theta) + (1 + \cos \theta / \sin \theta)$.

10-Second Method: If only sin & cos appear, put $\theta = 90^\circ$ ($\sin 90=1$, $\cos 90=0$). If tan/cot appear, put $\theta = 45^\circ$.

Live Example: Put $\theta=90^\circ$: $(1/1+0) + (1+0/1) = 1 + 1 = 2$. Option with $2 \csc(90^\circ) = 2(1) = 2$ is correct.

■ Geometry & Mensuration: 11-7 Divisibility Filter

Examiner Trap / Setup: Finding Volume or Area of Cylinder, Cone, Sphere, or Hemisphere.

10-Second Method: Formula contains $\pi = 22/7$ (multiple of 11). Test options for divisibility by 11.

Live Example: Check (Odd Sum - Even Sum) = 0 or 11. 3 out of 4 options fail instantly.

3. GENERAL INTELLIGENCE & REASONING: 10-SECOND ARSENAL

■ Syllogisms: The 100-50 Tick-Cross Method (Zero Venn Diagrams)

Examiner Trap / Setup: Drawing circles takes 30s. Use Income-Expense rules instead.

10-Second Method: All=100-50, No=100-100, Some=50-50, Some-Not=50-100. (1) Pos+Pos=Pos. (2) Pos+Neg=Neg. (3) 50 cannot become 100.

Live Example: Stmt: 'All Men are Smart (100-50)'. Concl: 'All Smart are Men (100-50)' → Smart is 50 in stmt, cannot be 100 in concl. FALSE in 2s.

■ Dice Problems: 1-Common-Face Clockwise Rule

Examiner Trap / Setup: Two dice views given with one common number (e.g., 3). Find opposite of 5.

10-Second Method: Write numbers clockwise starting from common face. Pairs in identical column positions are opposite.

Live Example: Dice 1: 3-5-1 | Dice 2: 3-2-6 → 5 is opposite 2, 1 is opposite 6, 3 is opposite 4. Solved in 4s.

■ Paper Folding & Cutting: The Single-Quadrant Filter

Examiner Trap / Setup: Complex paper cut unfolded into 4 quadrants. Finding the right option.

10-Second Method: Do NOT visualize the whole sheet. Pick ONE specific sharp notch in the cut quadrant and reflect it across the fold line.

Live Example: If triangle points LEFT in bottom-right, it MUST point RIGHT in bottom-left. Eliminate all options violating this.

■ Set Analogy: Smallest-Two-to-Largest Rule

Examiner Trap / Setup: Triplet given: (12, 16, 224). Find identical pattern option.

10-Second Method: Operate on the two smaller numbers (A & B) to form the largest (C): $(A+B) \times k$, or $(B-A)^2 \times k$, or (A^2+B^2) .

Live Example: $(12 + 16) = 28$. $28 \times 8 = 224$. Multiplier is 8. Match option satisfying $(A+B) \times 8 = C$.

4. ENGLISH COMPREHENSION: SPEED TACTICS

Question Topic	Common Error / Setter Pattern	Fastest Elimination Rule
Spotting Errors	Subject separated from verb by parenthetical clause (e.g., 'The quality of apples were...')	Ignore words between commas / 'along with' / 'of'. Match verb directly to the true root subject.
Inversion Traps	Hardly/Scarcely + when, No sooner + than.	Check conjunction pairs: <i>Hardly...when</i> (never than), <i>No sooner...than</i> (never when).
Para Jumbles (PQRS)	Full paragraph comprehension takes 60 seconds.	Find Mandatory Pairs: Locate Noun → Pronoun link (e.g. Newton in B, He in D → BD pair). Match options.
Active / Passive	Students read full sentences and get confused by tense.	Verb-Only Scan: Tense NEVER changes. V1 → is/are+V3, V2 → was/were+V3, has+V3 → has been+V3.
Direct / Indirect	Pronoun and tense shift confusion.	Backshift Scan: Said + Present → Past; Said + Past Simple → Past Perfect (had + V3).

5. GENERAL AWARENESS: 5-SECOND ELIMINATION PROTOCOL

General Awareness should take no more than **5 to 6 minutes for all 25 questions**. Use these rapid heuristics when recalling facts:

- **1. Extreme Words Elimination:** In Polity/Economics statements, options containing '*Always*', '*Never*', '*Completely*', '*Solely*' are FALSE in 90% of cases. Statements with '*May*', '*Can*', '*Some*' are usually TRUE.
- **2. Chronological Dynasty Anchors:** Sultanate dynasties follow **SKTSL** (Slave → Khilji → Tughlaq → Sayyid → Lodi). Eliminate out-of-order monuments or rulers immediately.
- **3. Classical Dance Exponent Keywords:** Maharaj/Kalka-Bindadin = *Kathak*; Reddys = *Kuchipudi*; Sapat/Pani = *Odissi*; Mohapatra = *Odissi*; Singh/Jhaveri Sisters = *Manipuri*.
- **4. Science Prefix Clues:** 'Hepato-' = Liver, 'Nephro-' = Kidney, 'Rhino-' = Nose, 'Osteo-' = Bones, 'Myo-' = Muscle. Match medical terms directly to organs.

6. MASTER OPTION ELIMINATION & GUESSING MATHEMATICS

Method	Core Principle	Example & Execution
Digital Root (Vedic Math)	Sum of digits reduced to single digit (cast out 9s). Invariant under +, -, ×.	$412 \times 318 = ? \rightarrow (4+1+2) \times (3+1+8) = 7 \times 12(3) = 21 \rightarrow 3$. Only the option with Digital Root 3 is correct.
Unit Digit Truncation	If options end in different digits, operate ONLY on last digits.	$487 \times 629 - 183 = ? \rightarrow (7 \times 9) - 3 = 63 - 3 = \text{ends in } 0$. Pick matching option in 2s.

The Expected Value Guessing Matrix (Negative Marking Engine):

In SSC CGL: Correct = **+2 Marks**, Incorrect = **-0.5 Marks**.

Scenario	Probability Calculation	Expected Value (EV)	Action Required
Eliminated 2 Options (Choose 1 of 2)	50% win (+2), 50% loss (-0.5)	+0.75 Marks per question	MANDATORY ATTEMPT (Huge Net Gain)
Eliminated 1 Option (Choose 1 of 3)	33% win (+2), 67% loss (-0.5)	+0.33 Marks per question	ATTEMPT (Positive Long-term EV)
Pure Blind Guess (1 of 4)	25% win (+2), 75% loss (-0.5)	+0.125 Marks	Skip unless overall attempts are < 70

7. THE 60-MINUTE EXAM HALL TIME ALLOCATION BLUEPRINT

Section	Ideal Time	Target Questions	Execution Strategy
1. Reasoning	14 – 15 Mins	23 – 25 / 25	Solve direct visual, syllogism & series first. Mark lengthy puzzles for round 2.
2. General Awareness	5 – 6 Mins	18 – 22 / 25	Binary recall: 5 seconds per question. Never dwell or calculate in GA.
3. English	8 – 10 Mins	23 – 25 / 25	Fast verb scanning on Active/Passive, Mandatory pairs for PQRS, Cloze test last.
4. Quantitative Aptitude	24 – 26 Mins	22 – 24 / 25	Apply value-putting, 11/7 divisibility filter, Digital Root, Unit digit.
5. Round 2 Buffer	3 – 4 Mins	Marked questions	Apply Guessing Matrix to 50/50 eliminated questions.